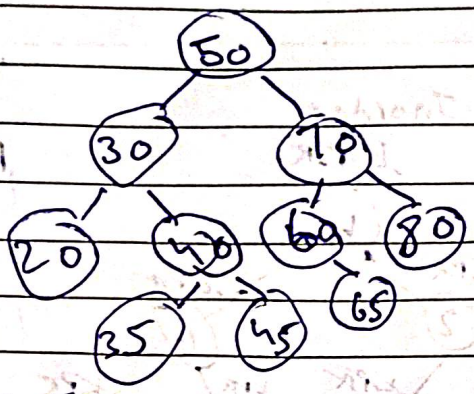
Section A.

Q1 Create BST

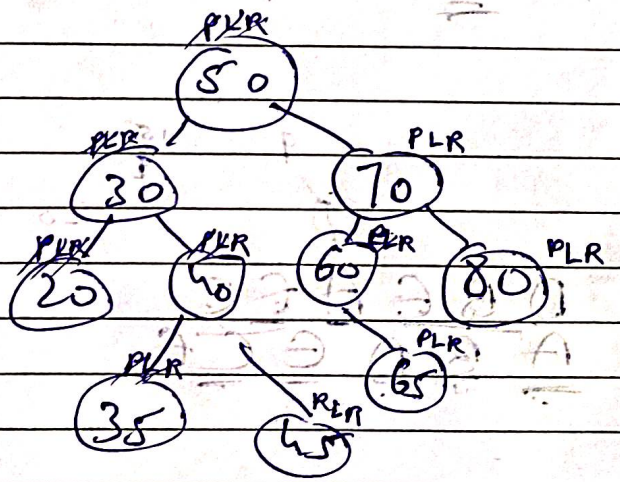
50 30 70 20 40 60 80 35 45 65



2) Depth, 0: 50
1: 30 70
2: 20 40 60 80
3.: 35 45 65

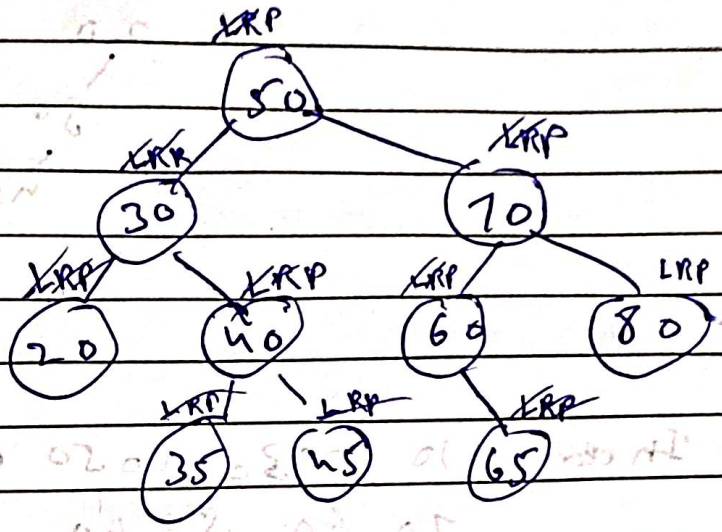
3) Inorder: 20, 30, 35, 40, 45, 50, 60, 65, 70

Preorder: PLR



Pre: 50, 30, 20, 40, 35, 45, 70, 60, 65, 80

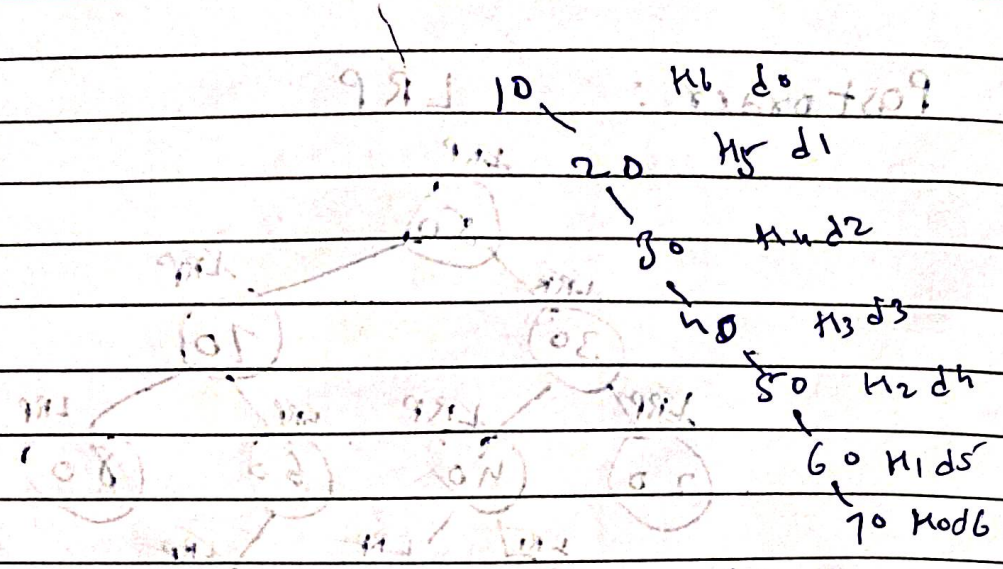
Postorder : LRP



Post : 20, 35, 45, 40, 30, 65, 60, 80, 70, 50

All leaf nodes - 35, 45, 65

6) Public Boolean searchNode (Node root, int key) {
if (root == null) return false;
else if (root == key) return true;
else if (root.data < key)
searchNode (root.right, key);
else
searchNode (root.prev, key);
return false;
}



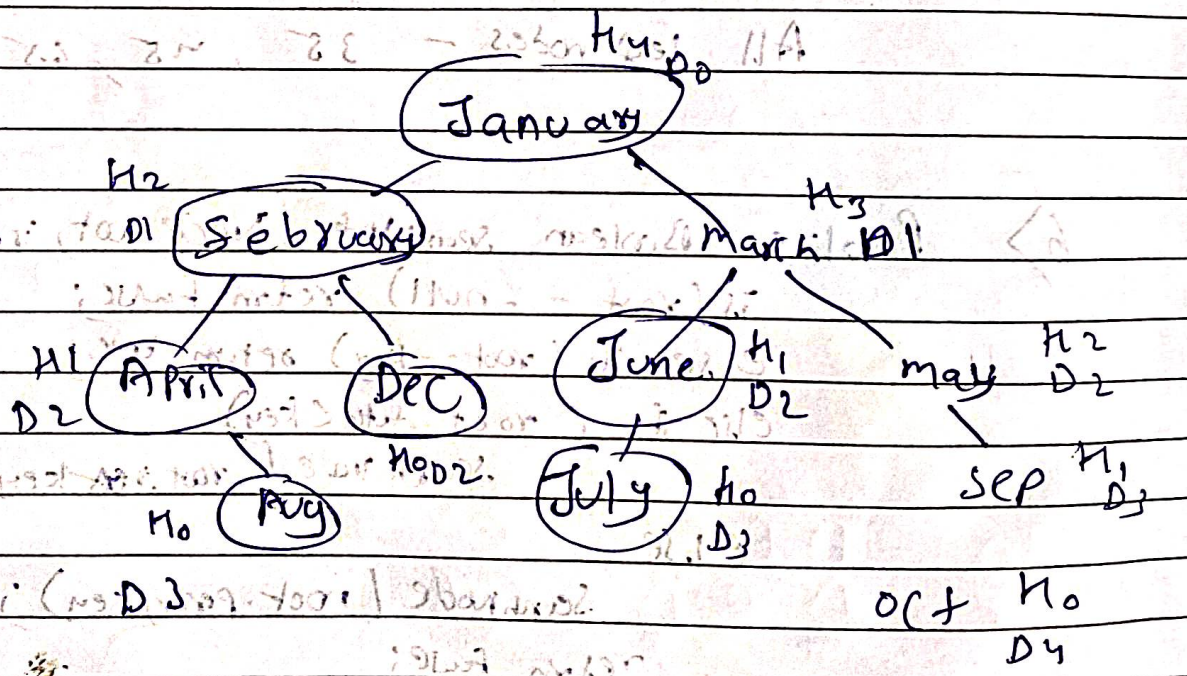
In order - 10, 20, 30, 40, 50, 60, 70

70, 60, 50, 40, 30, 20, 10

10, 20, 30, 40, 50, 60, 70

It is skewed because every node has at most one child.

Q3.



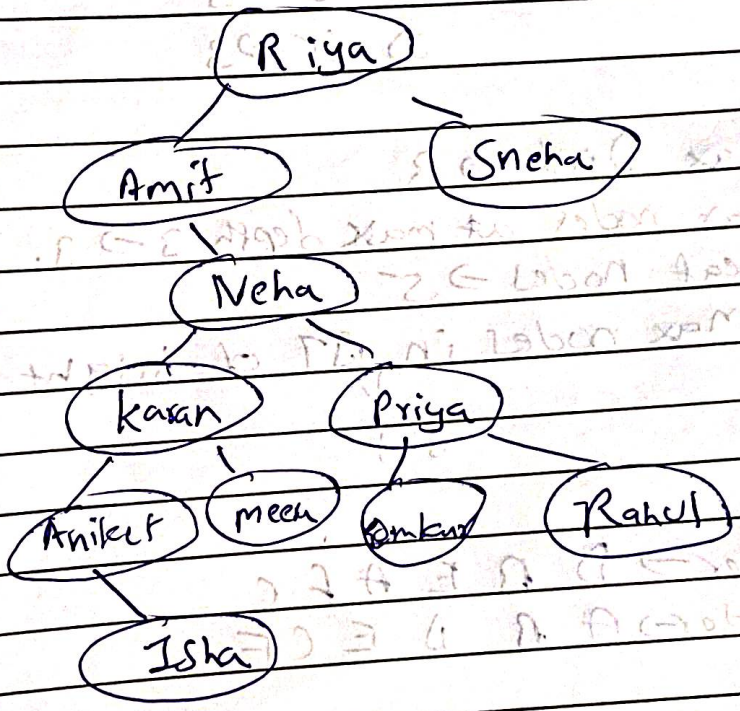
root → January

leaf nodes → Aug, Dec, July, October

Inorder → Aug, feb, dec, Jan July, Jun, ~~mar may~~
Preorder → Jan feb Apr Dec mar Jun July may sep oct
Post order → Aug Dec feb July June may march Jan.

Searchpaths →
① may :- July → January → March → febmay
② Aug :- July → Jan → March → Dec → Aug

Qs.



Inorder: Am, Ankit, Isha, karan meera Neha Omkar Priya Rahul Riya Sneha

Preorder: Riya Amit Neha karan Ankit Isha meera Priya omkar Rahul Sneha.

Postorder → Isha Ankit meera karan omkar Rahul Priya Neha Amit Sneha Riya

Search Path

↳ meera → Riya → Amit → Neha → Karan → meera

② Zoya → Riya → Sneha

Section B :-

Q6.

1) Depth & Height → 3

2) Depth of :- 1) 40 → 2

2) 45 → 3

3) 65 → 3

3) max Depth → 3

4) Max nodes at max depth 3 → 3.

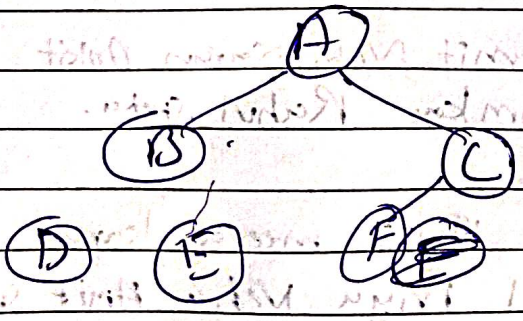
5) leaf nodes → 5

6) Max nodes in BT of height 4

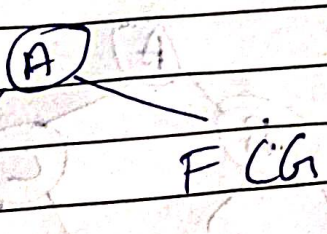
3^1

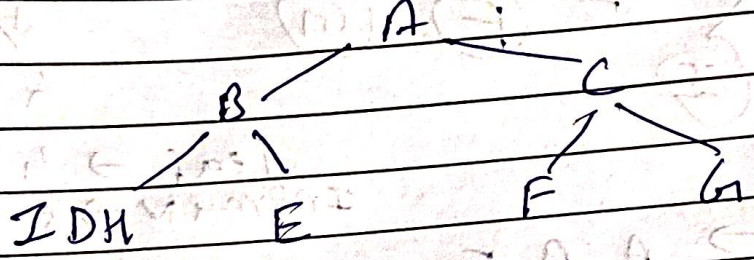
Q7. Inorder → D B E A F C

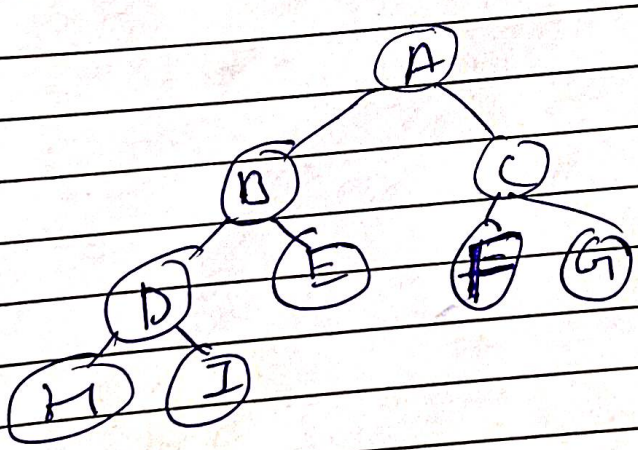
Preorder → A B D E C F



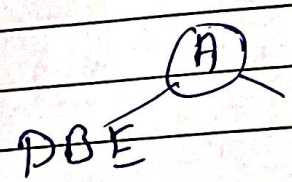
Q8) In order :- H D I B E A F C G
Post :- H D I B E F G C A

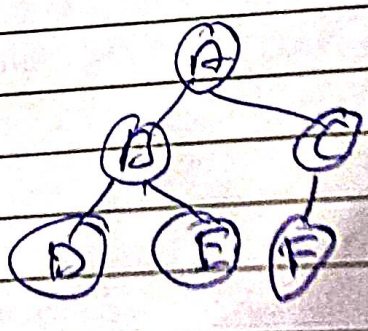




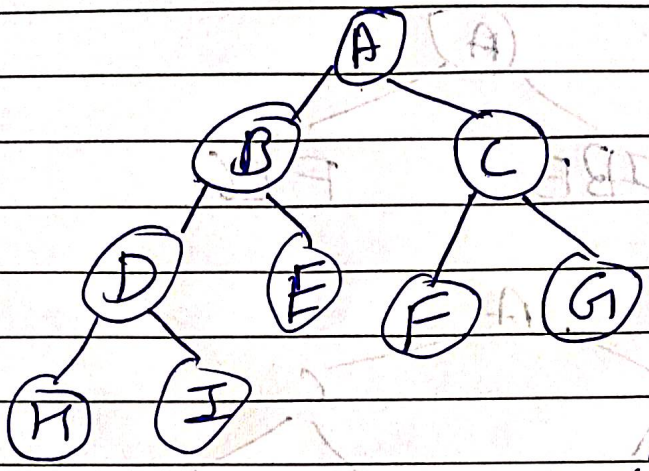


Q9) Inorder - D B E A F C
Post - D E B F C A





Q10) Inorder :- H D I B E A F C G
Postorder. H I D E B F G C A.

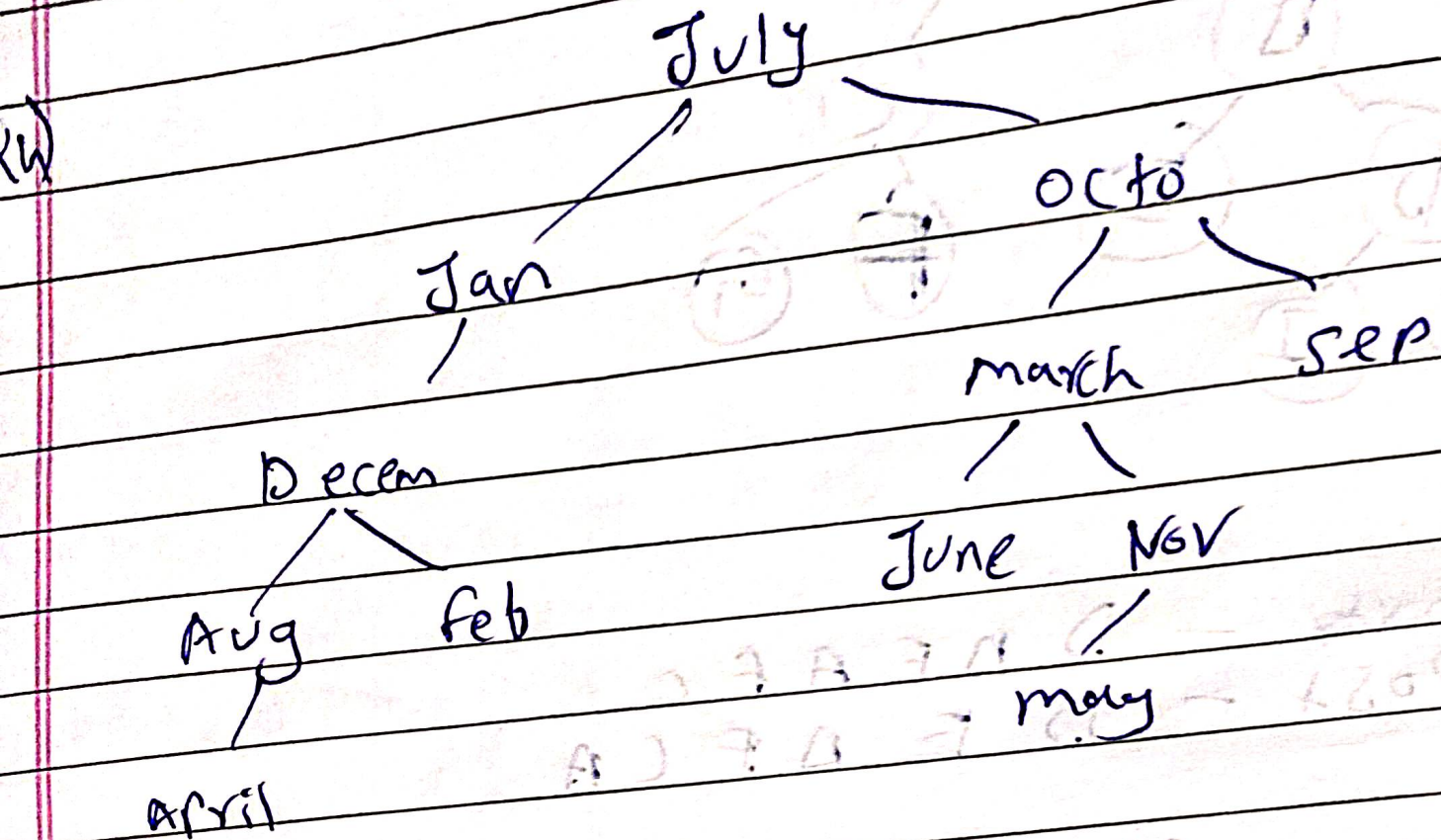


Depth & Height

leaf → 4
Internal → 5

Preorder → A B E D H I C F G

(W)



Inorder → April, Aug, Dec, Feb, Jan, July, March, May, Nov, oct, sept

Preorder → July Jan, Dec Aug April Feb, Dec marc June Nov may sep

Post → April Aug Feb Dec Jan June may Nov mar sep oct July